Lecture V – Continuous fields

Discretization

Continuous fields present a challenge for Decision Tree models because there are so many unique values. Splitting data on a unique field leads to too many partitions and an inability to calculate information gain. To overcome this challenge, we can utilize the discretization techniques discussed earlier. We can transform a unique field into a binary field by placing each record into one of two buckets. Determining the optimal split point can be done algorithmically:

1. First, order the data from lowest to highest
2. Determine candidate split points by finding the number half way between one record and the next record, for example, the candidate split point between 5 and 10 is 7.5
3. Iteratively partition the data on each of these candidate split points, calculating information gain each time
4. The candidate split point that leads to the greatest information gain will become the split point for that field and the records will be binned accordingly

This method can also be used to discretize a continuous target field.

Regression Trees

When the target of a decision tree is continuous, we have two solutions: we can discretize the target field or we can use a regression tree. To create a regression tree, we will follow these steps:

1. Select an input field as a candidate for the root split
2. Order the data according to this field
3. If the field is numeric, determine candidate split points
4. Iteratively partition the data on each of these candidate split points, calculating the average for each side of the partition and residual sum of square for each possible split
5. The candidate with the lowest residual sum of squares is the optimal split point
6. Repeat the process for other fields and select the field with the lowest residual sum of squares for the root split

Mean Absolute Error

We can judge the effectiveness of a regression tree by examining how much the prediction differs from the actual amount. Several measures can do this. To find the Mean Absolute Error, we can determine how much the predicted amount differs from the actual amount for each row of data. Then average of those differences. A lower Mean Absolute Error means the model is better at predicting the target field.

For example:

Actual	Prediction	Absolute Difference
5	6	1
8	6	2
		Mean Absolute Error
		1.5